

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TRANSMISSION CONTROL MODULE (TCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.
- If a failure condition is reported indicating the need to renew the transmission assembly on a vehicle that remains under manufacturer warranty, an authorisation request must first go through the warranty prior approval programme before any work is begun.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Transmission Control Module (TCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Diagnostics](#) (307-01A Automatic Transmission/Transaxle - Vehicles With: 8HP45 8-Speed Automatic Transmission AWD, Diagnosis and Testing) /

[Diagnostics](#) (307-01A Automatic Transmission/Transaxle - Vehicles With: 8HP45 8-Speed Automatic Transmission AWD, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B108E-02	Display - General signal failure	▪ Gear display signal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs. Road test the vehicle and check that the displayed gear matches the engaged gear. If the fault persists, install a new transmission control module
C1A88-64	High Pressure Switching Valve #1 - Signal plausibility failure	▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0219-86	Engine Overspeed Condition - Signal invalid	▪ Invalid data received from the engine control module	▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
P0560-21	System Voltage - Signal amplitude < minimum	▪ Battery/charging system fault ▪ Transmission control module power or ground circuit open circuit, high resistance	▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to

			the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance
P0562-1C	System Voltage Low - Circuit voltage out of range	▪ Battery/charging system fault ▪ Transmission control module power or ground circuit open circuit, high resistance	▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance
P0562-21	System Voltage Low - Signal amplitude < minimum	▪ Battery/charging system fault ▪ Transmission control module power or ground circuit open circuit, high resistance	▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance
P0563-22	System Voltage High - Signal amplitude > maximum	▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system

P0601-41	Internal Control Module Memory Checksum Error - General checksum failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-04	Control Module Processor - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-11	Control Module Processor - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-62	Control Module Processor - Signal compare failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-64	Control Module Processor - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-04	TCM Processor - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-06	TCM Processor - Algorithm based failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0613-11	TCM Processor - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-12	TCM Processor - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-13	TCM Processor - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-15	TCM Processor - Circuit short to battery or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-18	TCM Processor - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P061B-64	Internal Control Module Torque Calculation Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P062F-04	Internal Control Module EEPROM Error - System	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and

	internal failures		retest. If the fault persists, install a new transmission control module
P0641-64	Sensor Reference Voltage A Circuit/Open - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0642-21	Sensor Reference Voltage A Circuit Low - Signal amplitude < minimum	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0643-22	Sensor Reference Voltage A Circuit High - Signal amplitude > maximum	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0657-13	Actuator Supply Voltage A Circuit / Open - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0657-64	Actuator Supply Voltage A Circuit / Open - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0658-11	Actuator Supply Voltage A Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0659-12	Actuator Supply Voltage A Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0666-64	Control Module Internal Temperature Sensor "A" Circuit - Signal plausibility failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0667-01	PCM / ECM / TCM Internal Temperature Sensor A Range/Performance - General electrical failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0668-21	PCM / ECM / TCM Internal	Transmission control module failure	

Temperature
Sensor A Circuit
Low - Signal
amplitude <
minimum

NOTE:

Consider
environmental
conditions before
condemning the
transmission control
module

- Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0669-
22

PCM / ECM / TCM
Internal
Temperature
Sensor A Circuit
High - Signal
amplitude >
maximum

- Transmission control module failure

NOTE:

Consider
environmental
conditions before
condemning the
transmission control
module

- Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P06AC-
01

PCM / ECM / TCM
Internal
Temperature
Sensor B
Range/Performance
- General electrical
failure

- Transmission control module failure

NOTE:

Consider
environmental
conditions before
condemning the
transmission control
module

			Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P06AC-64	PCM / ECM / TCM Internal Temperature Sensor B Range/Performance - Signal plausibility failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P06AD-22	PCM / ECM / TCM Internal Temperature Sensor B Circuit Low - Signal amplitude > maximum	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P06AE-21	PCM / ECM / TCM Internal Temperature Sensor B Circuit High - Signal amplitude < minimum	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0700-02	Transmission Control System (MIL Request) - General signal failure	Transmission system fault	Investigate other DTCs first. Clear the DTCs and road test the vehicle. Re-read DTCs
P0700-83	Transmission Control System (MIL Request) - Value of signal protection calculation incorrect	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0702-98	Transmission Control System Electrical - Component or system over temperature	Insufficient transmission cooling	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Check the automatic transmission cooling system
P0710-04	Transmission Fluid Temperature Sensor A Circuit - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0710-	Transmission Fluid	Transmission control	Using the manufacturer

13	Temperature Sensor A Circuit - Circuit open	module failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0710-27	Transmission Fluid Temperature Sensor A Circuit - Signal rate of change above threshold	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0710-64	Transmission Fluid Temperature Sensor A Circuit - Signal plausibility failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0712-11	Transmission Fluid Temperature Sensor A Circuit	Transmission control module failure	

	Low - Circuit short to ground		NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0713-12	Transmission Fluid Temperature Sensor A Circuit High - Circuit short to battery	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0715-12	Turbine/Input Shaft Speed Sensor A Circuit - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0715-64	Turbine/Input Shaft Speed Sensor A	Turbine sensor signal plausibility failure	Using the manufacturer approved diagnostic

	Circuit - Signal plausibility failure		system, clear the DTCs and retest. Check datalogger signals - Turbine Speed (0x1E72) - Transmission Output Shaft Speed (0x1E68) - for plausibility. If the fault persists, install a new transmission control module
P0716-14	Turbine/Input Shaft Speed Sensor A Circuit Range/Performance - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0716-27	Turbine/Input Shaft Speed Sensor A Circuit Range/Performance - Signal rate of change above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0720-12	Output Shaft Speed Sensor Circuit - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0720-14	Output Shaft Speed Sensor Circuit - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0721-02	Output Shaft Speed Sensor Circuit Range/Performance - General signal failure	Output shaft speed sensor general signal failure	Check the transmission control module connector and the power/ground circuits to the transmission control module. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Clear the DTCs and retest. If the problem persists, renew the

			transmission control module as required
P0721-27	Output Shaft Speed Sensor Circuit Range/Performance - Signal rate of change above threshold	Signal rate of change above threshold	NOTE: This DTC may be set by an anti-lock braking system fault allowing the brakes to lock during braking Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0721-62	Output Shaft Speed Sensor Circuit Range/Performance - Signal compare failure	Output shaft speed sensor circuit signal compare failure	NOTES: - This DTC is not used in vehicles from 15MY onward - If this DTC is flagged, this may indicate that the vehicle has been fitted with an incorrect front and/or rear differential. Refer to the service instructions in Sections 205-02 & 205-03 before continuing with the diagnostic steps outlined below

			Check the transmission control module connector and the power/ground circuits to the transmission control module. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Clear the DTCs and retest. If the problem persists, renew the transmission control module as required
P0721-64	Output Shaft Speed Sensor Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	NOTE: If this DTC is flagged, this may indicate that the vehicle has been fitted with an incorrect front and/or rear differential. Refer to the service instructions in Sections 205-02 & 205-03 before continuing with the diagnostic steps outlined below Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P0725-81	Engine Speed Input Circuit - Invalid serial data received	Missing/invalid data from the engine control module	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
P0729-07	Gear 6 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid

			level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0730-00	Incorrect Gear Ratio - No sub type information	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0731-07	Gear 1 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0732-07	Gear 2 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0733-07	Gear 3 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved

			diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0734-07	Gear 4 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0735-07	Gear 5 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0736-07	Reverse Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0736-64	Reverse Incorrect Ratio - Signal plausibility failure	Automatic transmission failure	NOTE: This DTC is not used in vehicles from 15MY onward

			Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0741-07	Torque Converter Clutch Solenoid Circuit Performance/Stuck Off - Mechanical failures	Torque converter or automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new torque converter. Clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0745-11	Pressure Control Solenoid A - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0745-12	Pressure Control Solenoid A - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0745-13	Pressure Control Solenoid A - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0745-	Pressure Control	Transmission control	Using the manufacturer

14	Solenoid A - Circuit short to ground or open	module failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0745-64	Pressure Control Solenoid A - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0745-93	Pressure Control Solenoid A - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0745-94	Pressure Control Solenoid A - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0748-11	Pressure Control Solenoid A Electrical - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0748-12	Pressure Control Solenoid A Electrical - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0748-13	Pressure Control Solenoid A Electrical - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P076F-07	Gear 7 Ratio Incorrect - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P077E-02	Transmission Fluid Temperature Measurement System - Multiple Sensor Correlation - General signal failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0780-62	Shift Malfunction - Signal compare failure	Signal compare failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0780-93	Shift Malfunction - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0780-94	Shift Malfunction - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0795-04	Pressure Control Solenoid C - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P07AD-07	Transmission Friction Element "F" Stuck On - Mechanical failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P07D9-07	Gear 8 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07DC-07	Incorrect Shift from Gear 1 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07DD-07	Incorrect Shift from Gear 2 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07DE-07	Incorrect Shift from Gear 3 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the

			DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07DF-07	Incorrect Shift from Gear 4 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07E0-07	Incorrect Shift from Gear 5 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07E1-07	Incorrect Shift from Gear 6 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07E2-07	Incorrect Shift from Gear 7 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the

			vehicle. If the fault persists, install a new automatic transmission
P07E3-07	Incorrect Shift from Gear 8 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0801-94	Reverse Inhibit Control Circuit - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0810-04	Clutch Position Control Error - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Ensure the displayed gear matches the engaged gear. If the fault persists, install a new transmission control module
P084F-11	Park/neutral Switch Output Circuit - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-12	Park/neutral Switch Output Circuit - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-	Park/neutral Switch	Transmission control	Using the manufacturer

13	Output Circuit - Circuit open	module failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-14	Park/neutral Switch Output Circuit - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-29	Park/neutral Switch Output Circuit - Signal invalid	Park/neutral switch output circuit signal invalid	Check for correct output signal at transmission control module pin 1 to transmission plug (check in all positions). If the signal appears normal, then check wiring and connectors to the module. If no signal present, renew the transmission control module as required
P0850-02	Park/neutral Switch Input Circuit - General signal failure	Park/neutral switch output circuit failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and disengaging the parking lock several times. If fault persists, renew parklock components as necessary. If the fault persists, install a new transmission control module
P0850-29	Park/neutral Switch Input Circuit - Signal invalid	Starter inhibit signal invalid	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and disengaging the parking lock several times. If fault persists, renew parklock components as necessary. If the fault persists, install a

			new transmission control module
P0851-01	Park/neutral Switch Input Circuit Low - General electrical failure	Switch input circuit general signal failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and disengaging the parking lock several times. If fault persists, renew parklock components as necessary. If the fault persists, install a new transmission control module
P0852-01	Park/neutral Switch Input Circuit High - General electrical failure	Switch input circuit general signal failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and disengaging the parking lock several times. If fault persists, renew parklock components as necessary. If the fault persists, install a new transmission control module
P0942-07	Hydraulic Pressure Unit - Mechanical failures	NOTE: This DTC is only active with vehicles fitted with stop/start functionality Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0942-62	Hydraulic Pressure Unit - Signal compare failure		Perform ECO-start and monitor engine speed and engine/transmission start stop and position signals.

		NOTE: This DTC is only active with vehicles fitted with stop/start functionality Hydraulic pressure unit signal compare failure	Clear the DTCs and retest. If the problem persists, renew the transmission control module as required
P0965-64	Pressure Control Solenoid B Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0965-93	Pressure Control Solenoid B Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0965-94	Pressure Control Solenoid B Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0966-11	Pressure Control Solenoid B Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0966-12	Pressure Control Solenoid B Control Circuit Low - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0966-14	Pressure Control Solenoid B Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-13	Shift Solenoid A Control Circuit Range/Performance - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-64	Shift Solenoid A Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-93	Shift Solenoid A Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-94	Shift Solenoid A Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0973-11	Shift Solenoid A Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0973-18	Shift Solenoid A Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,

			install a new transmission control module
P0974-12	Shift Solenoid A Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0974-19	Shift Solenoid A Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-13	Shift Solenoid B Control Circuit Range/Performance - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-64	Shift Solenoid B Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-93	Shift Solenoid B Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-94	Shift Solenoid B Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0976-	Shift Solenoid B	Transmission control	Using the manufacturer

11	Control Circuit Low - Circuit short to ground	module failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0976-18	Shift Solenoid B Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0977-12	Shift Solenoid B Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0977-19	Shift Solenoid B Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0978-21	Shift Solenoid C Control Circuit Range/Performance - Signal amplitude < minimum	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0978-64	Shift Solenoid C Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0978-93	Shift Solenoid C Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0978-94	Shift Solenoid C Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0979-11	Shift Solenoid C Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0979-14	Shift Solenoid C Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0980-12	Shift Solenoid C Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0980-19	Shift Solenoid C Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0981-64	Shift Solenoid D Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0981-93	Shift Solenoid D Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,

			install a new transmission control module
P0981-94	Shift Solenoid D Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0982-11	Shift Solenoid D Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0982-14	Shift Solenoid D Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0982-18	Shift Solenoid D Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0983-12	Shift Solenoid D Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0983-19	Shift Solenoid D Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0984-	Shift Solenoid E	Transmission control	Using the manufacturer

64	Control Circuit Range/Performance - Signal plausibility failure	module failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0984-93	Shift Solenoid E Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0984-94	Shift Solenoid E Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0985-11	Shift Solenoid E Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0985-14	Shift Solenoid E Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0985-18	Shift Solenoid E Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0986-12	Shift Solenoid E Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0986-19	Shift Solenoid E Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P164C-62	Internal Control Module Start-Stop Performance - Signal compare failure	NOTE: This DTC is only active with vehicles fitted with stop/start functionality Transmission control module failure	Perform ECO-start and monitor engine speed and engine/transmission start stop and position signals. Clear the DTCs and retest. If the problem persists, renew the transmission control module as required
P1706-94	High Vehicle Speed Observed in Park - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P1707-72	Transfer Case Neutral or Park/Neutral Indication Circuit - Actuator stuck open	Actuator stuck open	- Ensure that emergency park release has not been pulled and is not sticking - Clear the DTCs and test drive the vehicle, engaging and disengaging the parking lock several times. If the problem persists, renew the transmission control module as required
P1707-74	Transfer Case Neutral or Park/Neutral Indication Circuit - Actuator slipping	- Emergency park release has been operated / is sticking - Parking lock mechanical failure	- Test the operation of the emergency park release - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P1707-77	Transfer Case Neutral or Park/Neutral Indication Circuit - Commanded position not reachable	Commanded position not reachable	- Ensure that emergency park release has not been pulled and is not sticking - Clear the DTCs and test drive the vehicle, engaging and disengaging the parking lock several times. If the problem persists, renew the transmission control module as required
P1707-94	Transfer Case Neutral or Park/Neutral Indication Circuit - Unexpected operation	- Emergency park release has been operated / is sticking - Parking lock mechanical failure	- Test the operation of the emergency park release - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P177A-07	Transmission Friction Element A or B - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177B-07	Transmission Friction Element A or C - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177C-07	Transmission Friction Element A or D - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid

			level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177D-07	Transmission Friction Element A or E - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177E-07	Transmission Friction Element B or C - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177F-07	Transmission Friction Element B or D - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P178A-07	Transmission Friction Element B or E - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved

			diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P178B-07	Transmission Friction Element C or D - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P178C-07	Transmission Friction Element C or E - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P178D-07	Transmission Friction Element D or E - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P215B-62	Vehicle Speed / Output Shaft Speed Correlation - Signal compare failure	- Incorrect final drive ratio fitment - Incorrect ABS software	- Check for the correct final drive ratio fitment and rectify as required - Using the manufacturer approved diagnostic system, check the anti-lock brake system control

			module for related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, reconfigure the anti-lock brake system control module with the latest level software
P258F-02	Torque Management Request Output Signal - General signal failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2638-67	Torque Management Feedback Signal A Range/Performance - Signal incorrect after event	Torque intervention from engine control module	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
P2700-07	Transmission Friction Element A Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2701-07	Transmission Friction Element B Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2702-	Transmission	Automatic	Refer to the relevant

07	Friction Element C Apply Time Range/Performance - Mechanical failures	transmission failure	section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2703-07	Transmission Friction Element D Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2704-07	Transmission Friction Element E Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2711-94	Unexpected Mechanical Gear Disengagement - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2713-04	Pressure Control Solenoid D - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P2722-04	Pressure Control Solenoid E - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2757-93	Torque Converter Clutch Pressure Control Solenoid Control Circuit Perf or Stuck Off - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2757-94	Torque Converter Clutch Pressure Control Solenoid Control Circuit Perf or Stuck Off - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2759-11	Torque Converter Clutch Pressure Control Solenoid Electrical - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2759-12	Torque Converter Clutch Pressure Control Solenoid Electrical - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2761-13	Torque Converter Clutch Pressure Control Solenoid Control Circuit / Open - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2762-64	Torque Converter Clutch Pressure Control Solenoid Control Circuit Range/Perf - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P2784-02	Input/Turbine Speed Sensor A/B Correlation - General signal failure	▪ Sensor general signal failure, multiple input speed signal faults	▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
P2787-4B	Clutch Temperature Too High - Over temperature	▪ Clutch over temperature	▪ Allow the transmission to cool, clear the DTCs and retest. Check transmission oil level and check for excessive gear changes during normal driving. Check paddle switch for intermittent operation and renew as required
P2787-98	Clutch Temperature Too High - Component or system over temperature	▪ Component or system over temperature	▪ Allow the transmission to cool, clear the DTCs and retest. Check transmission oil level and check for excessive gear changes during normal driving. Check paddle switch for intermittent operation ▪ Check the transmission oil level and quality, refer to the relevant section of the workshop manual. Clear the DTCs and test drive the vehicle using all gears in drive and reverse. Clear the DTCs and retest. If the problem persists, or if any other gear ratio related DTCs are logged, renew the transmission
P2793-92	Gear Shift Direction Circuit - Performance or	▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and

	incorrect operation		retest. If the fault persists, install a new transmission control module
P2793-94	Gear Shift Direction Circuit - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-83	High Speed CAN Communication Bus - Value of signal protection calculation incorrect	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-86	High Speed CAN Communication Bus - Signal invalid	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related

			DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	Missing message from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0073-00	Control Module Communication Bus "A" Off - No sub type information	High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0121-87	Lost Communication With Anti-Lock Brake System (ABS) Control Module - Missing message	Missing message from the anti-lock brake system control module	Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
U0300-57	Internal Control Module Software Incompatibility - Invalid/incomplete software component	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U0401-81	Invalid Data Received from ECM/PCM A - Invalid serial data received	Missing/invalid data from the engine control module	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U0401-82	Invalid Data Received from	Missing/invalid data from the engine	Using the manufacturer approved diagnostic

	ECM/PCM A - Alive/sequence counter incorrect/not updated	control module	system, check the engine control module for related DTCs and refer to the relevant DTC index
U0401-83	Invalid Data Received from ECM/PCM A - General checksum failure	Missing/invalid data from the engine control module	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U0401-87	Invalid Data Received from ECM/PCM A - Missing message	Missing/invalid data from the engine control module	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U0415-82	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Alive/sequence counter incorrect / not updated	Missing/invalid data from the anti-lock brake system control module	Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
U0415-83	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - General checksum failure	Missing/invalid data from the anti-lock brake system control module	Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
U2101-56	Control Module Configuration Incompatible - Invalid/incomplete configuration	Transmission control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the transmission control module with the latest level software. Check and update the car configuration file as necessary. Clear the DTCs and retest
U3000-56	Control Module - Invalid/incomplete configuration	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and

			retest. If the fault persists, install a new transmission control module
U3000-9A	Control Module - Component or system operating conditions	Engine control module fault	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U3001-94	Control Module Improper Shutdown - Unexpected operation	- Transmission control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance - Refer to the relevant section of the workshop manual and test the battery and charging system